

Mathletes Formulas

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1. Polynomials

1.1. Linear Equations

For polynomials of degree one, the variable m is used to refer to the slope of the lines drawn by their graphs. Parallel lines have equivalent slopes, while perpendicular lines have slopes that are reciprocals of one another ($m \perp \frac{1}{m}$).

Slope-Intercept

$$y = mx + b$$

Point-Slope

$$y - y_1 = m(x - x_1)$$

Slope

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-Point

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

1.2. Quadratics

Quadratic Equations

Standard Form $ax^2 + bx + c$

Let $D = b^2 - 4ac$,

Vertex Form $a(x - h)^2 + k$

1. $D > 0 \implies 2$ real solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. $D = 0 \implies 1$ real solution

3. $D < 0 \implies$ no real solutions

Minimum and Maximum Values

Consider the function $f(x) = ax^2 + bx + c$ with vertex $(-\frac{b}{2a}, f(-\frac{b}{2a}))$

1. $a > 0 \implies f$ has a *minimum* at $x = -\frac{b}{2a}$. The minimum value is $f(-\frac{b}{2a})$.

2. $a < 0 \implies f$ has a *maximum* at $x = -\frac{b}{2a}$. The maximum value is $f(-\frac{b}{2a})$.

1.3. General Polynomials

Let n be a non-negative integer and let $a_n, a_{n-1}, \dots, a_2, a_1, a_0$ be real numbers such that $a_n \neq 0$. The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of x with degree n .

Linear Factorization Theorem

If $f(x)$ is a polynomial of degree n , where $n > 0$, then $f(x)$ has precisely n linear factors

$$f(x) = a_n(x - c_1)(x - c_2)\dots(x - c_n)$$

where $c_1, c_2, \dots, c_n$ are complex numbers.

Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of f has the form $\frac{p}{q}$, where p and q have no common factors other than 1, and

p = a factor of the constant term a_0

q = a factor of the leading coefficient a_n

Conjugate Pairs

Let f be a polynomial function with *real coefficients*. If $a + bi$, where $b \neq 0$, is zero of f , the complex conjugate $a - bi$ is also a zero of f .

For polynomials with *integer coefficients*, $a + \sqrt{b}$ being a zero implies $a - \sqrt{b}$ is also a zero.

Roots

Let f be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

$$\text{The sum of all roots} = -\frac{a_n}{a_{n-1}}$$

$$\text{The product of all roots} = (-1)^n \frac{a_0}{a_n}$$

1.4. Rational Functions

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0}$$

where $N(x)$ and $D(x)$ have no common factors.

1. The graph of f has *vertical asymptotes* at the zeros of $D(x)$.
2. The graph of f has at most one *horizontal asymptote* determined by comparing the degrees of $N(x)$ and $D(x)$.
 - a. $n < 0 \implies$ the line $y = 0$ is the horizontal asymptote.
 - b. $n = 0 \implies$ the line $y = \frac{a_n}{b_n}$ is the horizontal asymptote.
 - c. $n > 0 \implies$ there is no horizontal asymptote.

Suppose $N(x)$ and $D(x)$ had some common factor $(x - c)$. The graph of f would then have a hole at $(c, f(c))$

2. Trigonometry

2.1. Functions

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\csc x = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\sin x = \frac{y}{1} = y \quad \csc x = \frac{1}{y}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\sec x = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\cos x = \frac{x}{1} = x \quad \sec x = \frac{1}{x}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}}$$

$$\cot x = \frac{\text{adjacent}}{\text{opposite}}$$

$$\tan x = \frac{y}{x} \quad \cot x = \frac{x}{y}$$

Function	Domain	Range	Period
sin	$\mathbb{R}$	$[-1, 1]$	2π
cos	$\mathbb{R}$	$[-1, 1]$	2π
tan	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq 0\}$	$\mathbb{R}$	π
csc	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq \frac{1}{2}\}$	$(-\infty, -1] \cup [1, \infty)$	2π
sec	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq 0\}$	$(-\infty, -1] \cup [1, \infty)$	2π
cot	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq \frac{1}{2}\}$	$\mathbb{R}$	π

Table 1: Functions

Function	Domain	Range
$\sin^{-1} / \arcsin$	$[-1, 1]$	$[-90^\circ, 90^\circ]$
$\cos^{-1} / \arccos$	$[-1, 1]$	$[0^\circ, 180^\circ]$
$\tan^{-1} / \arctan$	$\mathbb{R}$	$[-90^\circ, 90^\circ]$
$\csc^{-1} / \operatorname{arccsc}$	$(-\infty, -1] \cup [1, \infty)$	$[-90^\circ, 90^\circ]$
$\sec^{-1} / \operatorname{arcsec}$	$(-\infty, -1] \cup [1, \infty)$	$[0^\circ, 180^\circ]$
$\cot^{-1} / \operatorname{arccot}$	$\mathbb{R}$	$[0^\circ, 180^\circ]$

Table 2: Inverse Functions

	0°	30°	45°	60°	90°	180°	270°	360°
sin	0	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
cot	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef	0	undef	0

Table 3: Common Values

2.2. Identities

Radians and Degrees

$$n^\circ = \frac{\pi}{180} n \text{ rad}$$

$$m \text{ rad} = \left(\frac{180}{\pi} m \right)^\circ$$

Reciprocals

$$\csc x = (\sin x)^{-1} \quad \sin x = (\csc x)^{-1}$$

$$\sec x = (\cos x)^{-1} \quad \cos x = (\sec x)^{-1}$$

$$\cot x = (\tan x)^{-1} \quad \tan x = (\cot x)^{-1}$$

Pythagorean

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Even/Odd

$$\begin{aligned}\sin(-x) &= -\sin x & \csc(-x) &= -\csc x \\ \cos(-x) &= \cos x & \sec(-x) &= \sec x \\ \tan(-x) &= -\tan x & \cot(-x) &= -\cot x\end{aligned}$$

Sum-to-Difference

$$\begin{aligned}\sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y \\ \cos(x \pm y) &= \cos x \cos y \pm \sin x \sin y \\ \tan(x \pm y) &= \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}\end{aligned}$$

Half-Angle

$$\begin{aligned}\sin\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 - \cos x}{2}} * \\ \sin\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 + \cos x}{2}} * \\ \tan\left(\frac{x}{2}\right) &= \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} *\end{aligned}$$

Double-Angle

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x \\ \cos 2x &= \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x \\ \tan 2x &= \frac{2 \tan x}{1 - \tan^2 x}\end{aligned}$$

* where you consider the angle's location to determine the sign

Sum-to-Product

$$\begin{aligned}\sin x + \sin y &= 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \sin x - \sin y &= 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \\ \cos x + \cos y &= 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \cos x - \cos y &= -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)\end{aligned}$$

$$\begin{aligned}\tan x \pm \tan y &= \frac{\sin(x \pm y)}{\cos x \cos y} \\ \text{where } \frac{x}{90^\circ} \bmod 2 &\neq 1 \\ \cot x \pm \cot y &= \frac{\sin(y \pm x)}{\sin x \sin y} \\ \text{where } \frac{x}{90^\circ} \bmod 2 &\neq 2\end{aligned}$$

Product-to-Sum

$$\begin{aligned}\sin x \sin y &= \frac{\cos(x-y) - \cos(x+y)}{2} \\ \cos x \cos y &= \frac{\cos(x+y) + \cos(x-y)}{2} \\ \sin x \cos y &= \frac{\sin(x+y) + \sin(x-y)}{2}\end{aligned}$$

Co-Function

$$\begin{aligned}\sin(90 - x) &= \cos x & \cos(90 - x) &= \sin x \\ \sec(90 - x) &= \csc x & \csc(90 - x) &= \sec x \\ \tan(90 - x) &= \cot x & \cot(90 - x) &= \tan x\end{aligned}$$

2.3. Laws**Law of Sines**

$$\begin{aligned}\frac{\sin A}{a} &= \frac{\sin B}{b} = \frac{\sin C}{c} \\ \text{Area} &= \frac{1}{2} ab \sin A\end{aligned}$$

Law of Cosines

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A \\ \cos A &= \frac{b^2 + c^2 - a^2}{2bc}\end{aligned}$$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{1}{2}(A-B)\right)}{\tan\left(\frac{1}{2}(A+B)\right)}$$

3. Graphs and Geometry

3.1. Lines and Points

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

3.2. Shapes

Circles

$$(x - m)^2 + (y - n)^2 = r^2$$

$$A = \pi r^2$$

$$C = 2\pi r$$

Regular Polygons

$$\text{Sum of Interior Angles} = 180(n - 2)^\circ$$

$$\text{Area} = \frac{1}{2} \cdot \text{Apothem} \cdot \text{Perimeter}$$

Volumes

$$V_{\text{Cylinder}} = \pi r^2 h$$

$$V_{\text{Cone}} = \frac{1}{3} \pi r^2 h$$

$$V_{\text{Sphere}} = \frac{4}{3} \pi r^3$$

Triangle Similarity

Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if

SSS

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

SAS

$$\frac{AB}{DE} = \frac{AC}{DF} \text{ and } \angle A = \angle D$$

AA

$$\angle A = \angle D \text{ and } \angle B = \angle E$$

3.3. Graph Manipulation

Transformation	Conditions	Description
$g(x) = f(x \pm c)$	(+)	shift left c units
	(-)	shift right c units
$g(x) = f(x) \pm c$	(+)	shift left c units
	(-)	shift right c units
$g(x) = cf(x)$	$c < 0$	vertical compress by a factor of c
	$c > 0$	vertical stretch by a factor of c
$g(x) = f(cx)$	$c < 0$	horizontal stretch by a factor of c

Transformation	Conditions	Description
	$c > 0$	horizontal compress by a factor of c
$g(x) = -f(x)$		reflect about the x-axis
$g(x) = f(-x)$		reflect about the y-axis

4. Miscellaneous

4.1. Inverse Functions

Let f and g be two function such that

$$f(g(x)) = x \text{ for all } x \text{ in the domain of } g \qquad g(f(x)) = x \text{ for all } x \text{ in the domain of } f$$

Under these conditions, the function g is the *inverse function* of the function f (and vice-versa). The function g can then be denoted as f^{-1} . So,

$$f(f^{-1}(x)) = x \qquad f^{-1}(f(x)) = x$$

The domain of f must be equal to the range of f^{-1} , and the range of f must be equal to the domain of f^{-1} .

4.2. Complex Numbers

Let a and b be real numbers. The number $a + bi$ is a *complex number* written in *standard form*. The number a is the *real part* and the number bi is the *imaginary part* of the complex number.

When $b = 0$, the number $a + bi$ is a real number. When $b \neq 0$, the number $a + bi$ is an *imaginary number*. A number of the form bi , where $b \neq 0$, is a *pure imaginary number*.

$$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^n = i^{n \bmod 4}$$

4.3. Exponential and Logarithmic Relationships

Exponential		Logarithmic	
$a^0 = 1$	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	$\log_a 1 = 0$	$\log ab^n = n \log ab$
$a^m \cdot a^n = a^{m+n}$	$(a^m)^n = a^{mc \cdot n}$	$\log_a a = 1$	$\log a^n b = \frac{1}{n} \log ab$
$\frac{a^m}{a^n} = a^{m-n}$	$a^{\frac{1}{n}} = \sqrt[n]{a}$	$\log x = \log_{10} x$	$\log_n ab = \log_n a + \log_n b$
$(ab)^n = a^n \cdot b^n$	$a^m = a^n \implies m = n$	$\ln x = \log_e x$	$\log_n \frac{a}{b} = \log_n a - \log_n b$
		$y = \log_b x \implies b^y = x$	$\log_n a = \log_n b \implies a = b$
		$a^{\log_a b} = b$	